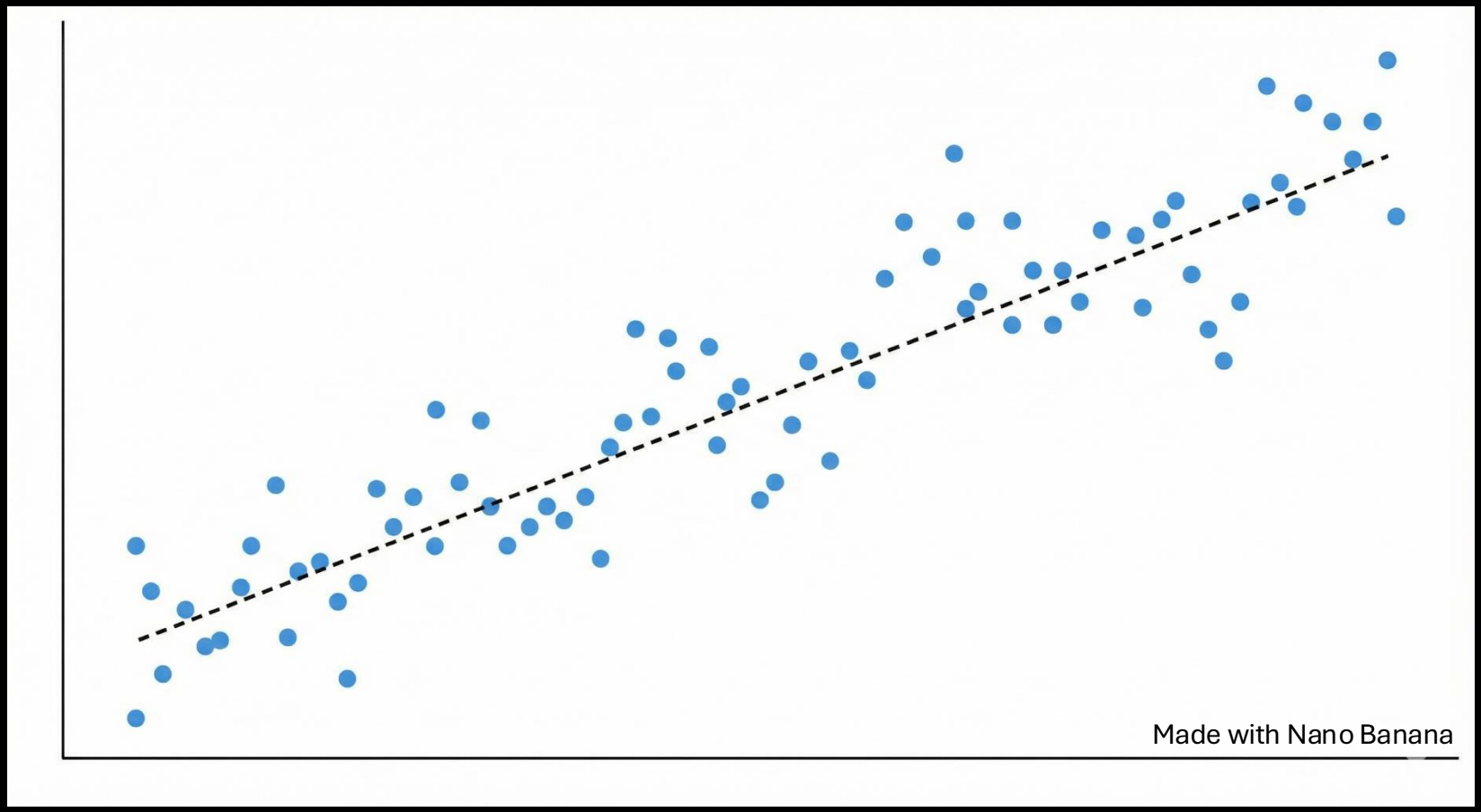
Module 2: Basic Machine Learning

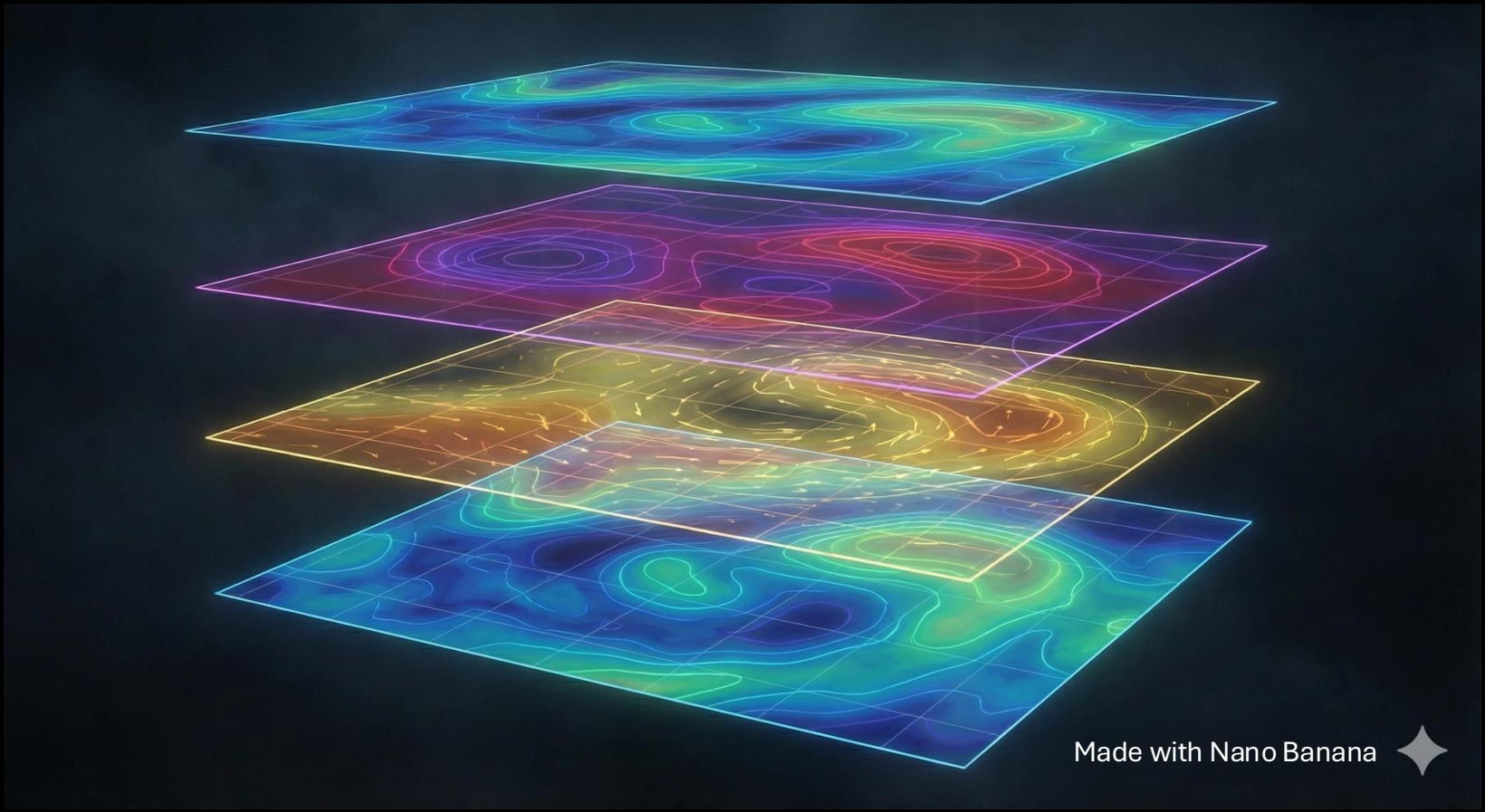
Machine learning doesn’t have to be complicated.



Find y = mx + b so that the sum of the squared distances of all data points (x) from the line (y) are minimized. Thus, minimize L2 loss, i.e., mean squared error (MSE). In other words, find a line that fits the data.

You probably did this in middle school or high school on a TI-89.

But it can be



Find the hyperparameter set that optimizes a generalized fit to multi-dimensional data that also varies through time.

You probably have never done this.

ML has been used in weather prediction for *decades*...

Model Output Statistics (MOS)

Predictors:

- NWP Model Forecasts
- Prior Observations
- Geoclimatic Data

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OBV   N  N  N BR  N  N  N HZ  N  N  N BR  N  N  N BR  N BR FG  N HZ
```

Historically, a multi-linear regression

What makes a good problem for machine learning?

Data rich

What do we have in atmospheric science:

- Surface stations (decades; global but biased toward populated, wealthy regions)
- Weather balloons (decades; global but biased toward populated, wealthy regions)
- Satellite data (years of high-resolution multispectral data)
    - Can be used to remotely derive multiple quantities in true global manner
- Buoy data (locally concentrated in areas of interest like equatorial Pacific and near shore)
- Ship data (locally concentrated on ship paths)
- Aircraft data (locally concentrated on ship paths)

This data is ingested (assimilated) into an atmospheric model of past weather and climate, known as reanalysis.

What data do we use? Usually reanalysis, although newer models experimenting with ML DA.

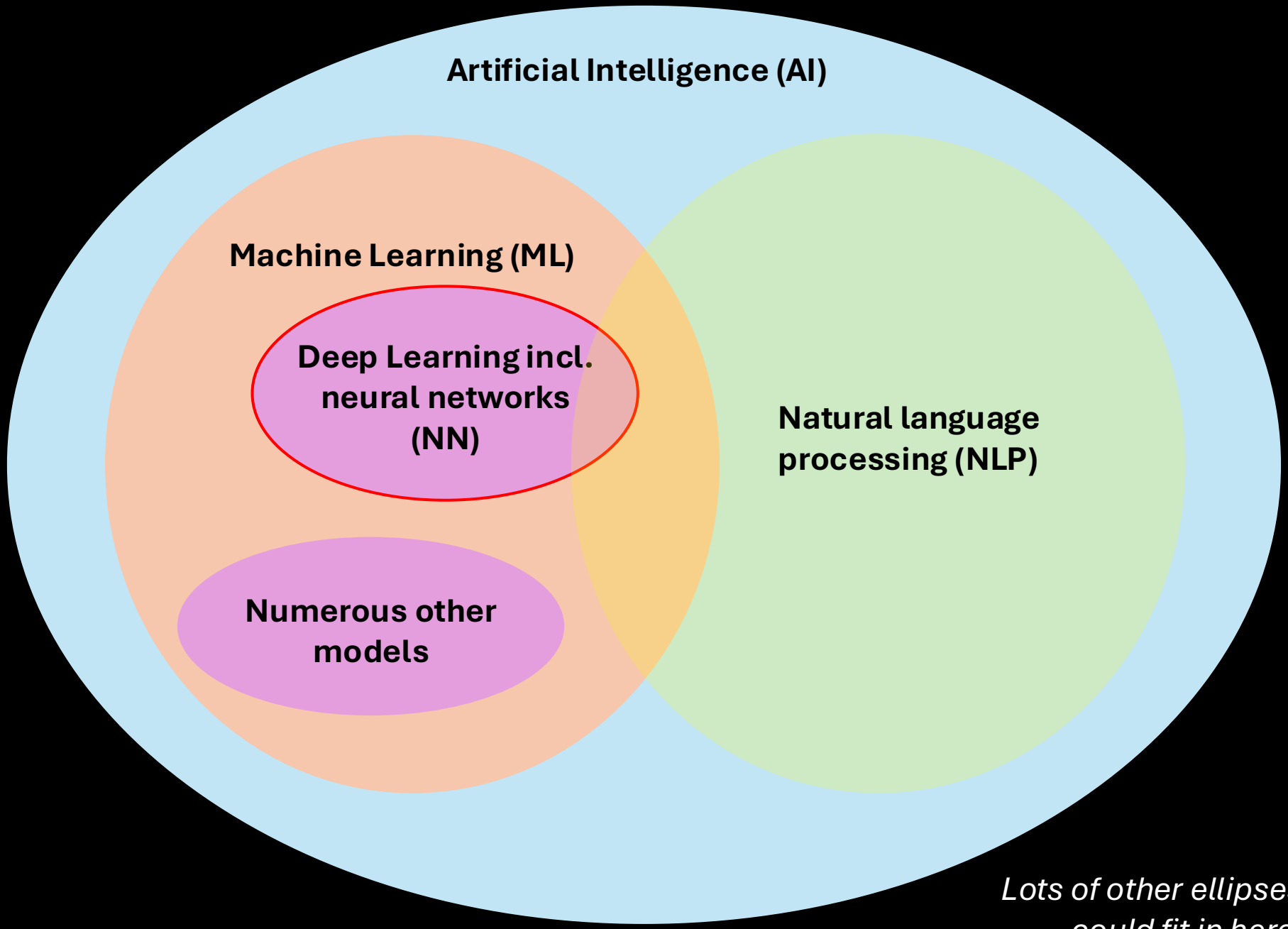
REMEMBER: Reanalysis is a model output that ingests any available observations.

Reanalysis has parameterizations:
- Microphysics
- Cumulus
- Surface
- Radiation
- Turbulence/boundary layer

But is repeatedly nudged toward observations so it doesn't stray too far from the truth.

Large-scale dynamic fields like wind are often good.
Local fields like precipitation depend on parameterization and often have large errors and biases. This can indirectly impact other variables like relative humidity.
Other important variables like temperature depend on height. Upper-level usually well constrained, but near-surface values can suffer biases.

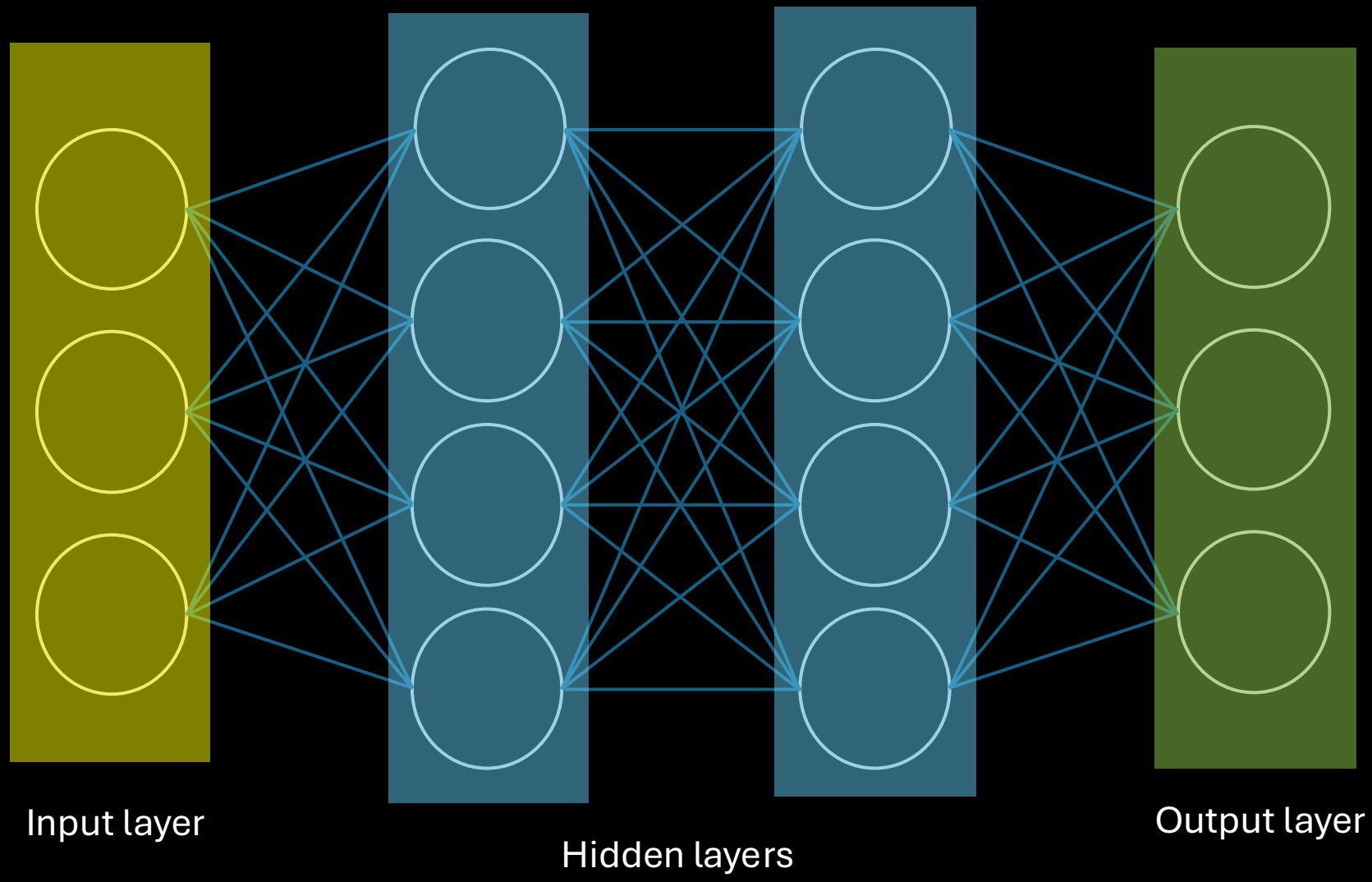
Many of today's ML weather models use reanalysis as the truth, but the truth still contains errors!



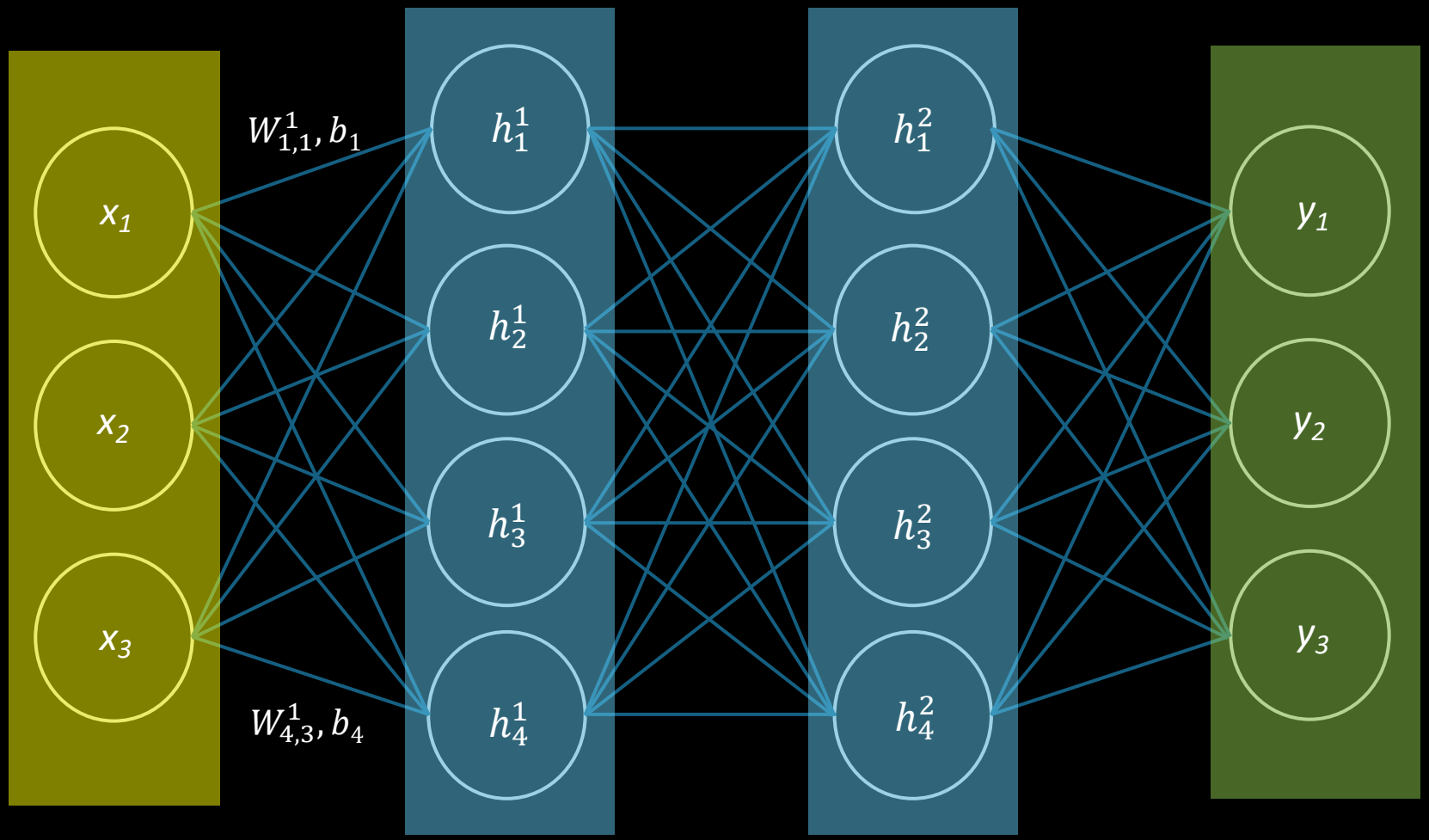
Lots of other ellipses could fit in here.

How does a neural network function?

A neural network is essentially a mathematical framework for transforming input into output—except *instead of using derived equations, the transformations are learned using data*.



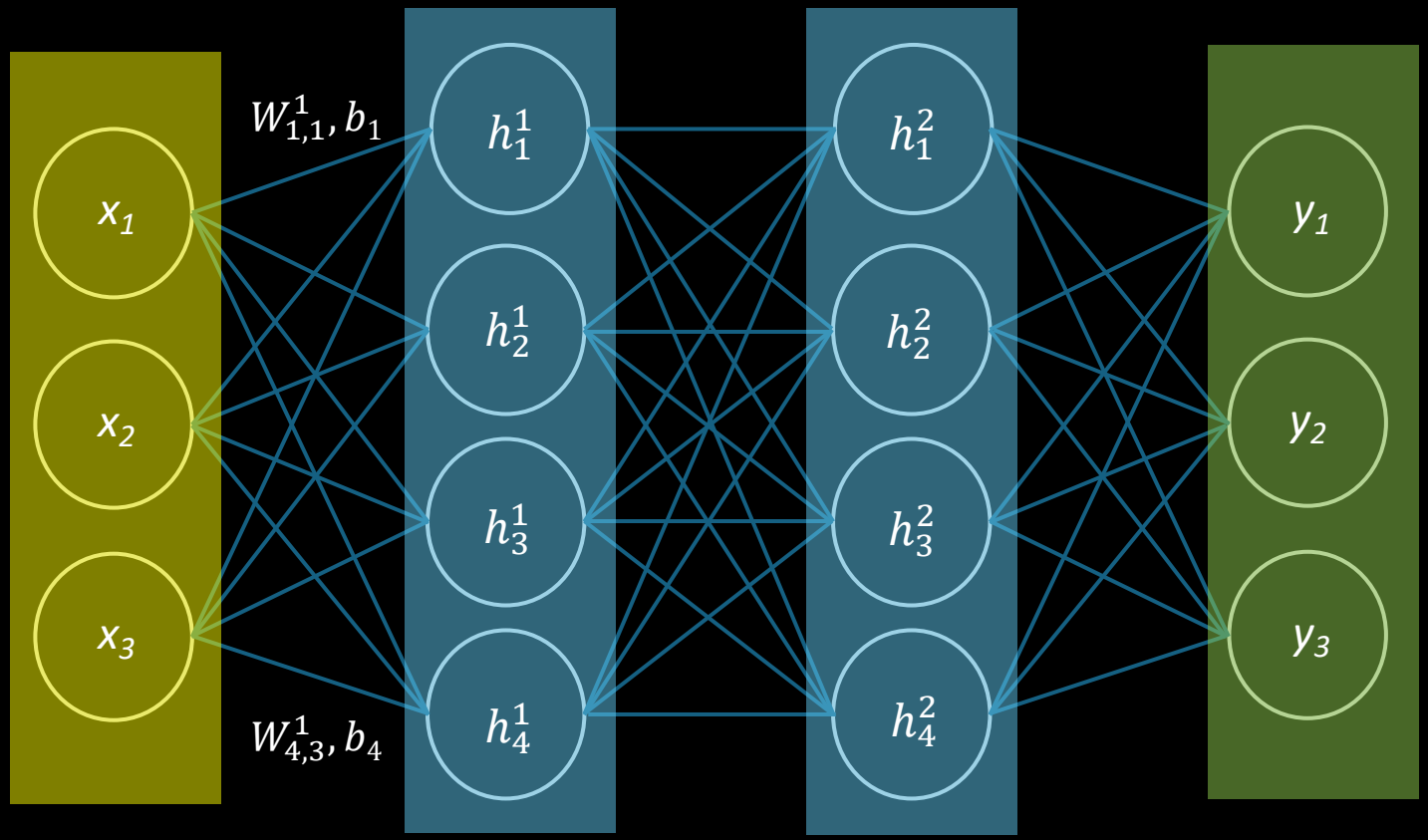
How does a neural network function?



$$h^1_j = \sigma\left(\sum_{i=1}^{n} W^1_{j,i}\, x_i + b^1_j\right) \qquad h^2_j = \sigma\left(\sum_{i=1}^{n} W^2_{j,i} h^1_j + b^2_j\right) \qquad y_j = \sigma\left(\sum_{i=1}^{n} W^3_{j,i} h^2_j + b^3_j\right)$$

How does a neural network function?

$$h = \sigma(Wx + b)$$



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h: hidden layers (new representation of data)

W: weights per neuron (learned)

b: biases (learned per neuron)

x: input layers (normalized data)

σ: Activation function (imposes nonlinearity)

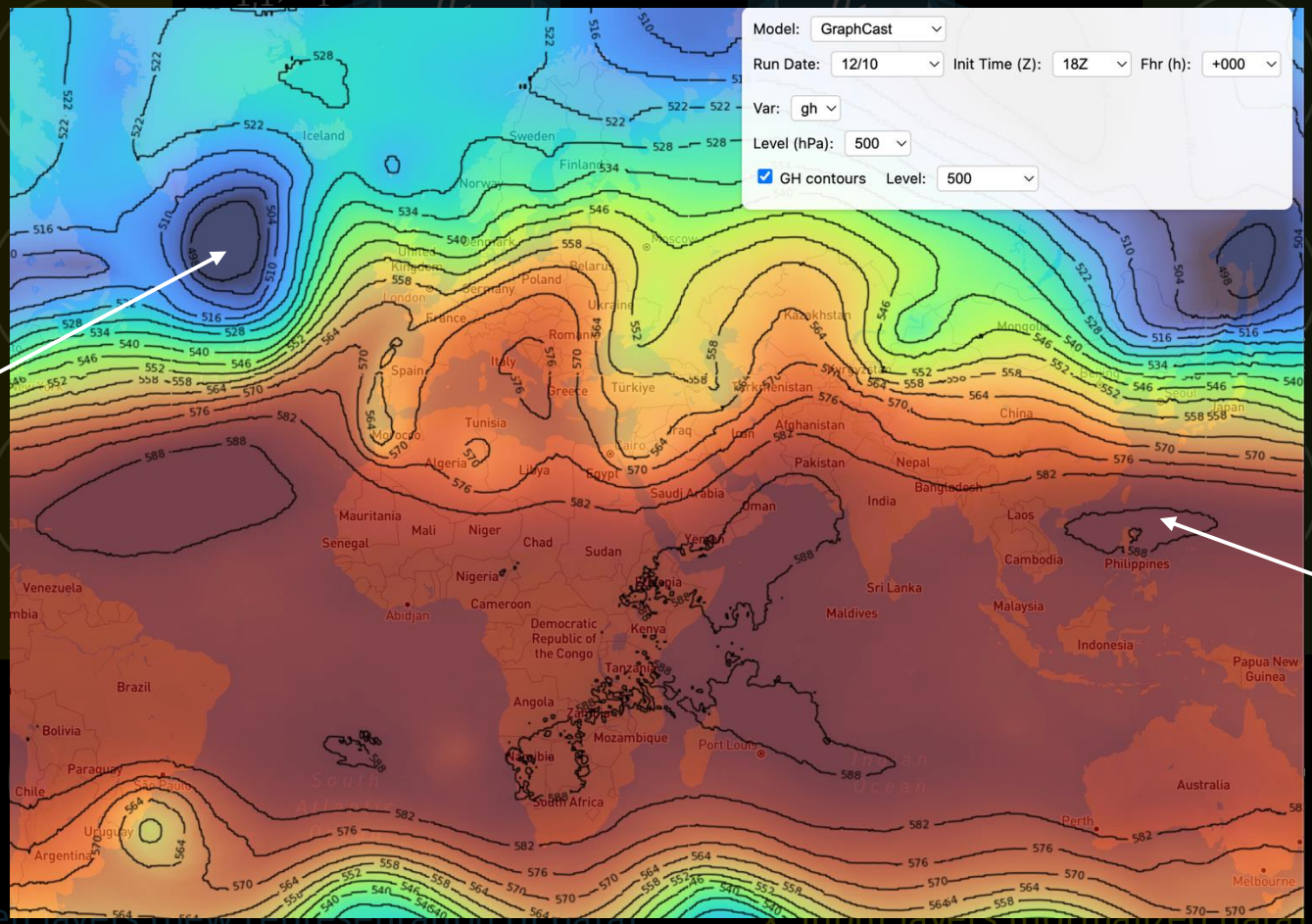
But…

How does a neural network function?

$$h = \sigma(Wx + b)$$

This is a fully connected neural network (FCNN). That means every neuron in any layer talks to every neuron in each layer before and after it.

Do dynamics here…

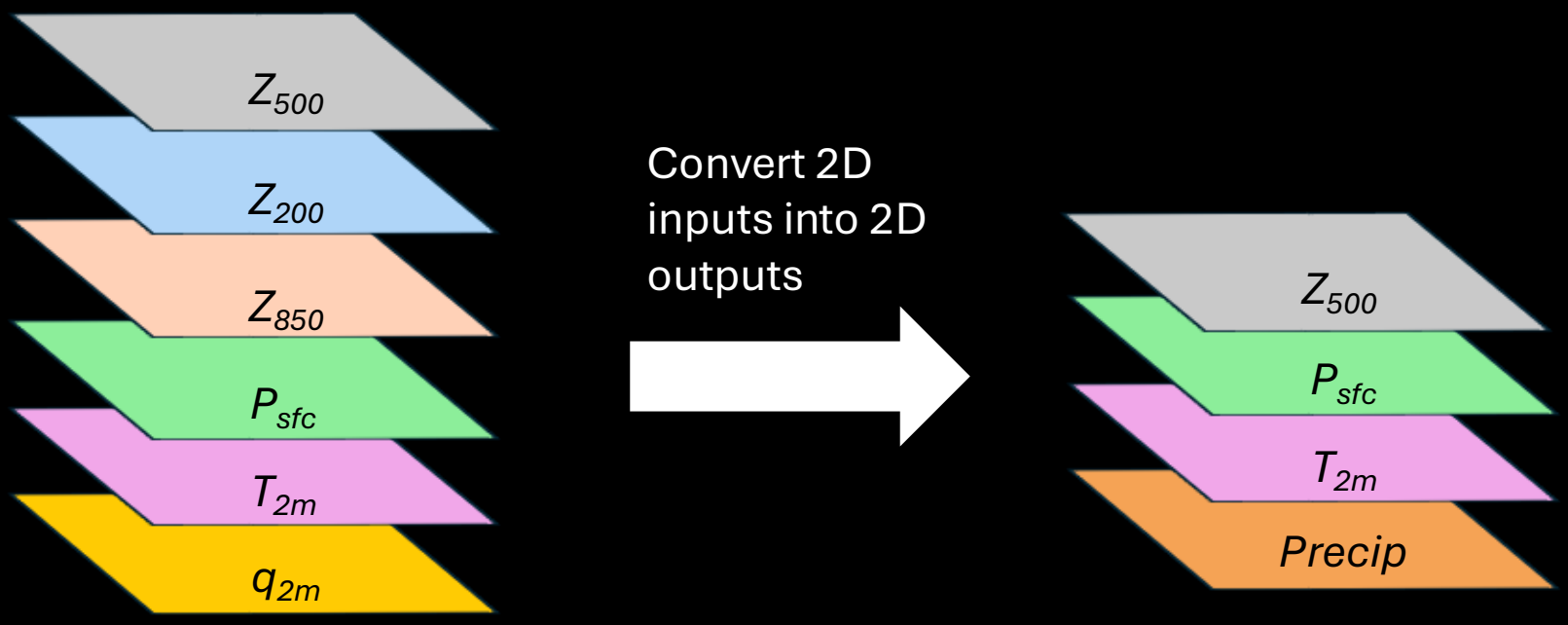


…impact the forecast here 6 hours later?

Imagine every variable at every grid point in a discretized model is a neuron. Using a FCNN would be nonsensical.

Convolutional neural networks (CNNs) take advantage of *spatial* information.

Interactive CNN Visualization: https://poloclub.github.io/cnn-explainer/

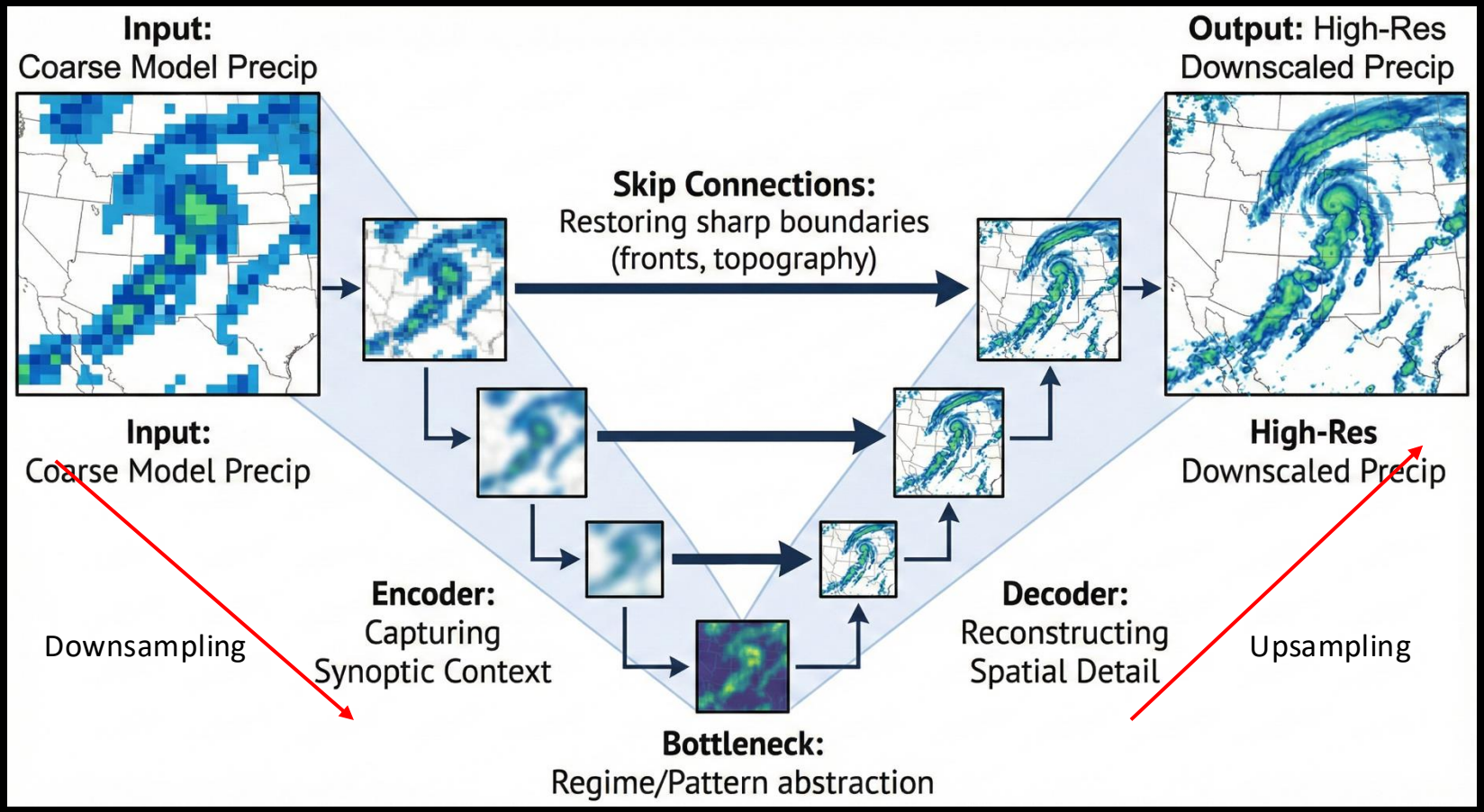


Inputs (examples shown)

Outputs (could be same as or different than inputs)

CNNs can have various different types of architecture.

Below is a common type of CNN: A U-Net architecture. This is example of image sharpening.



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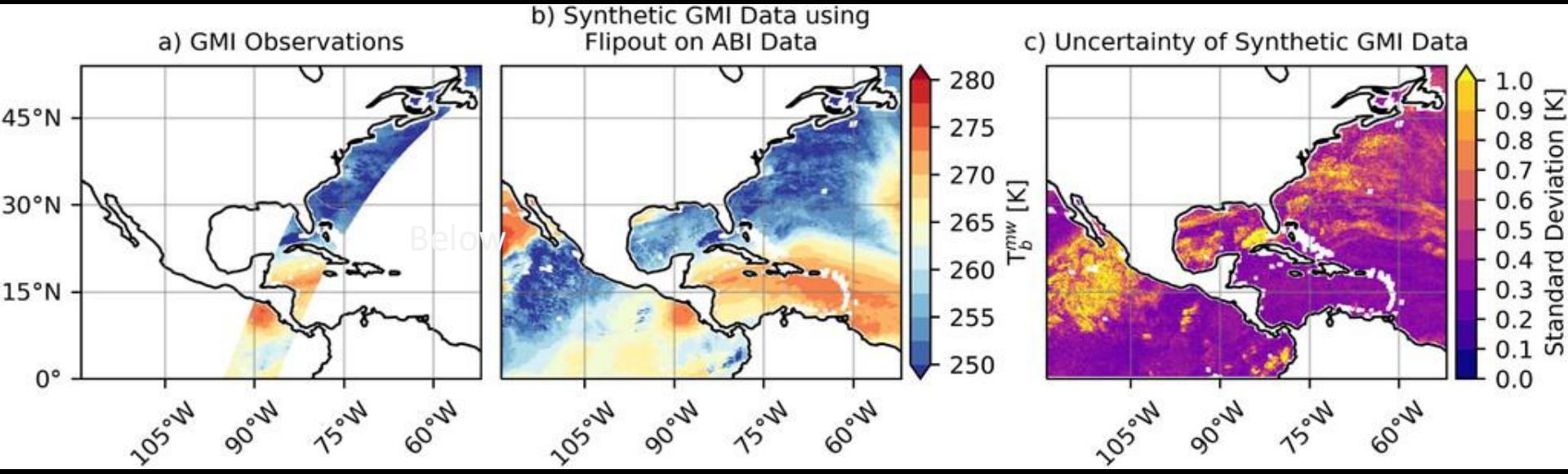
Bayesian ResNet architecture: Emulate microwave satellite data using geostationary IR data with uncertainties predicted



Single observed MW swath

Full scene of predicted MW data

Standard deviation of predictions

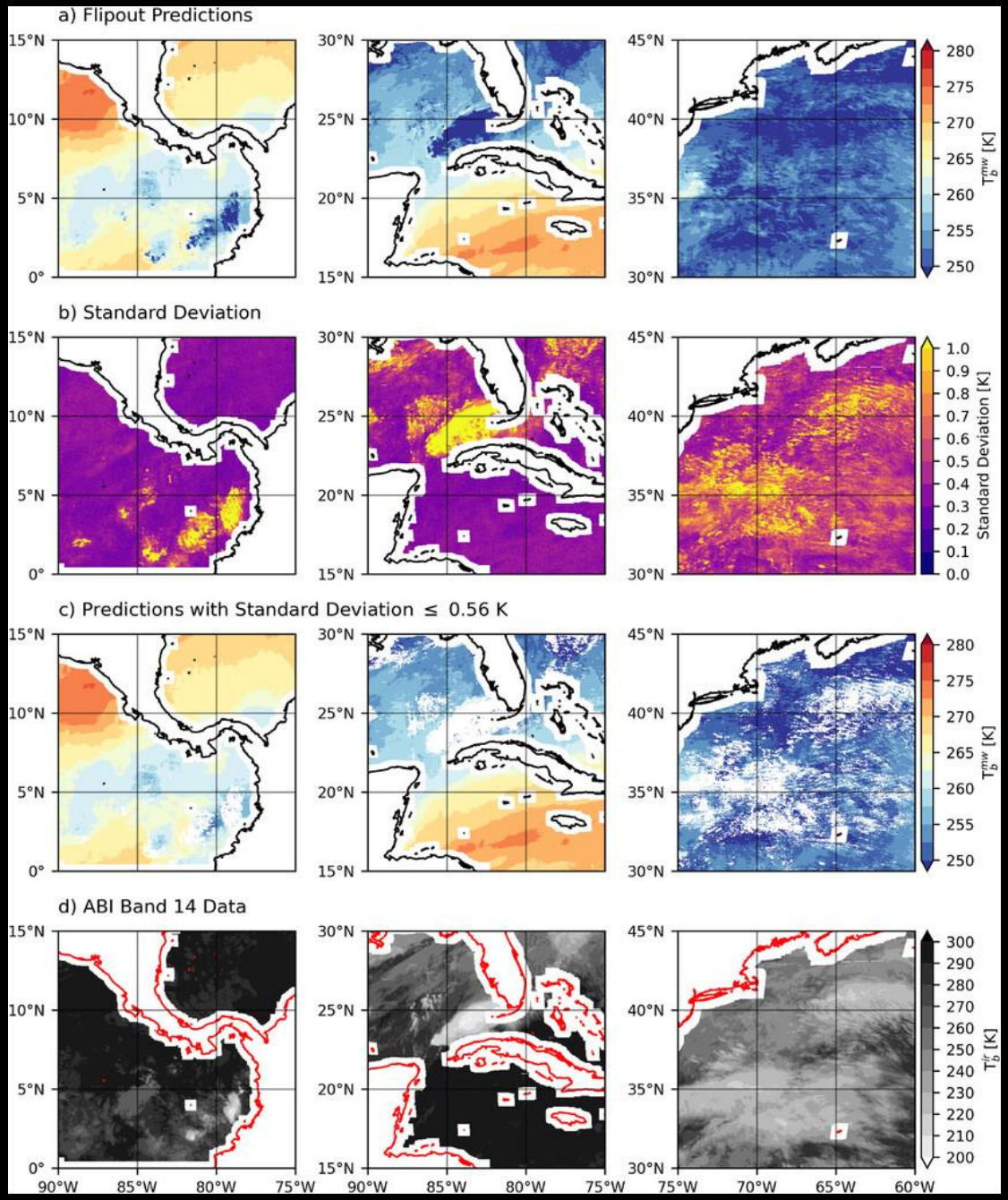
This is also a CNN.

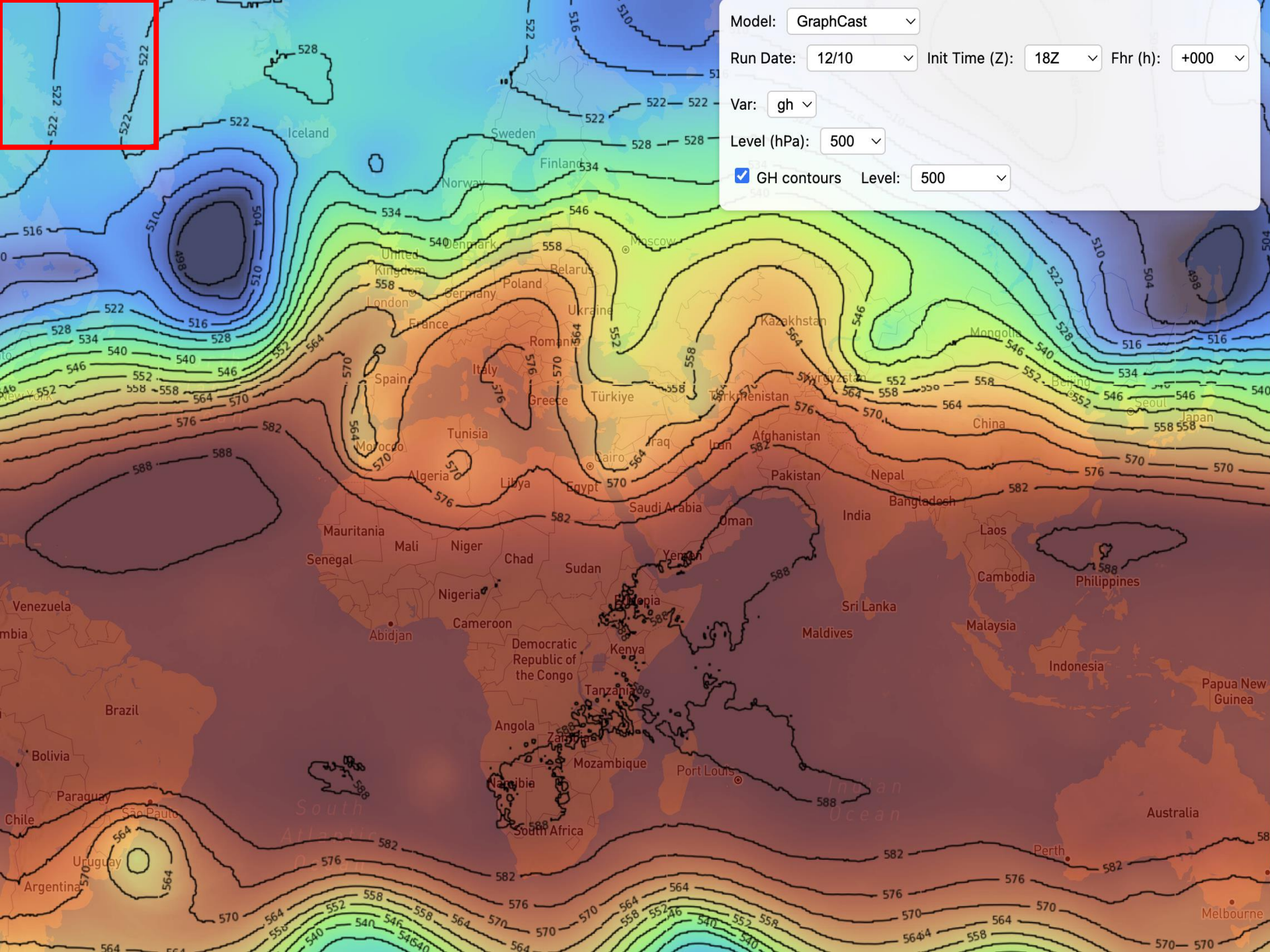
Predictions

Uncertainty metric

Mask out uncertain regions

Part of the input tensor

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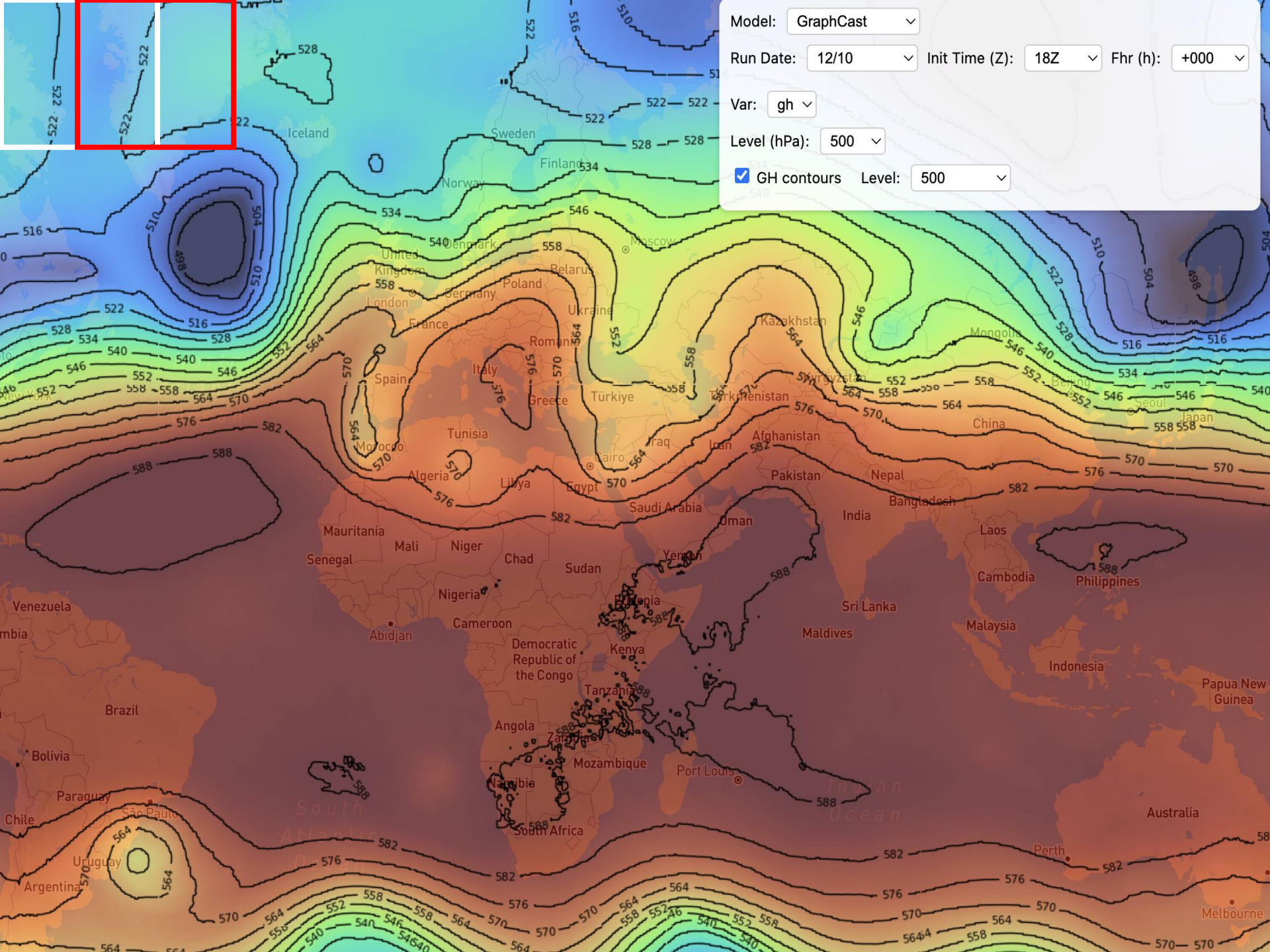
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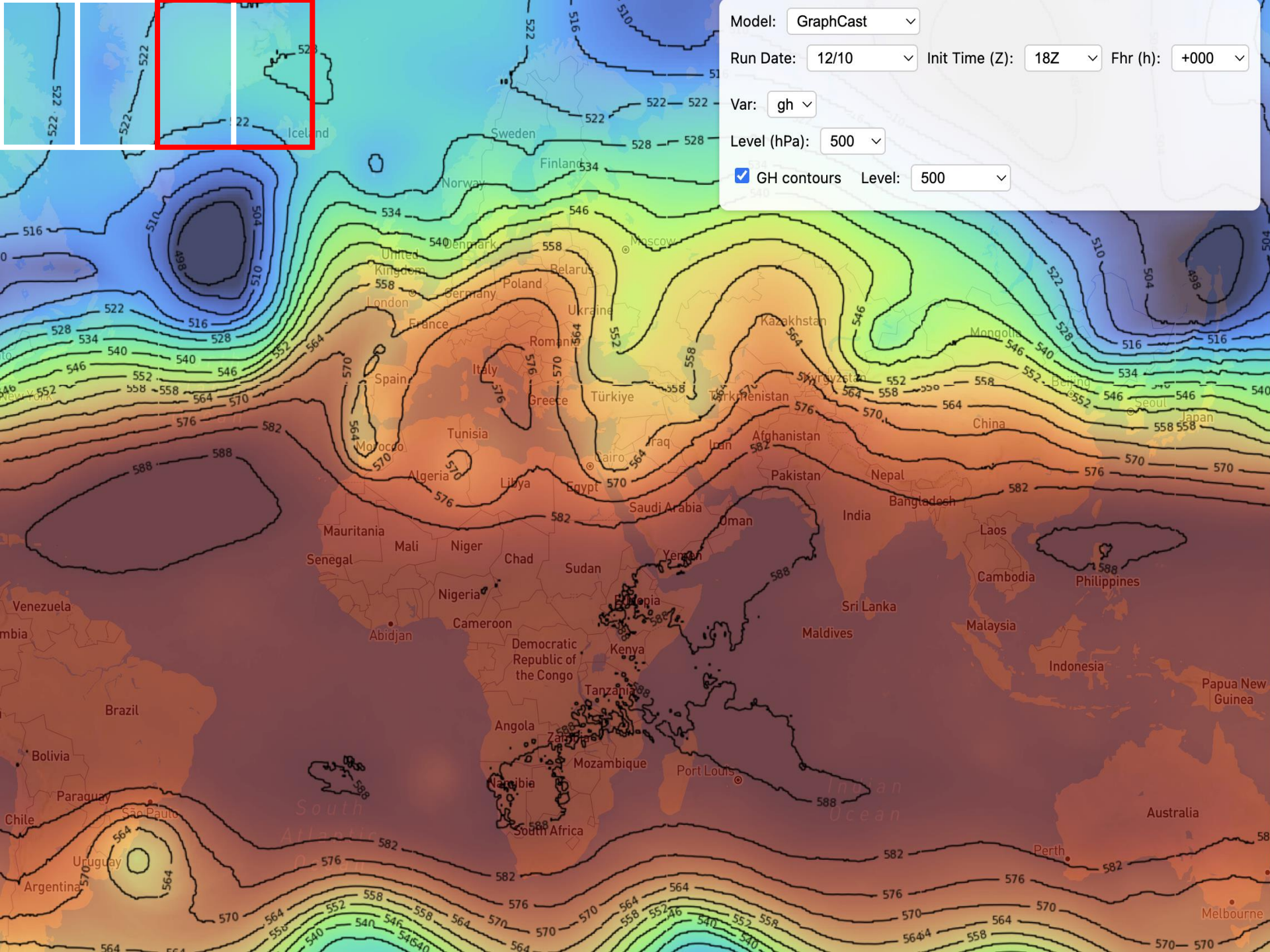
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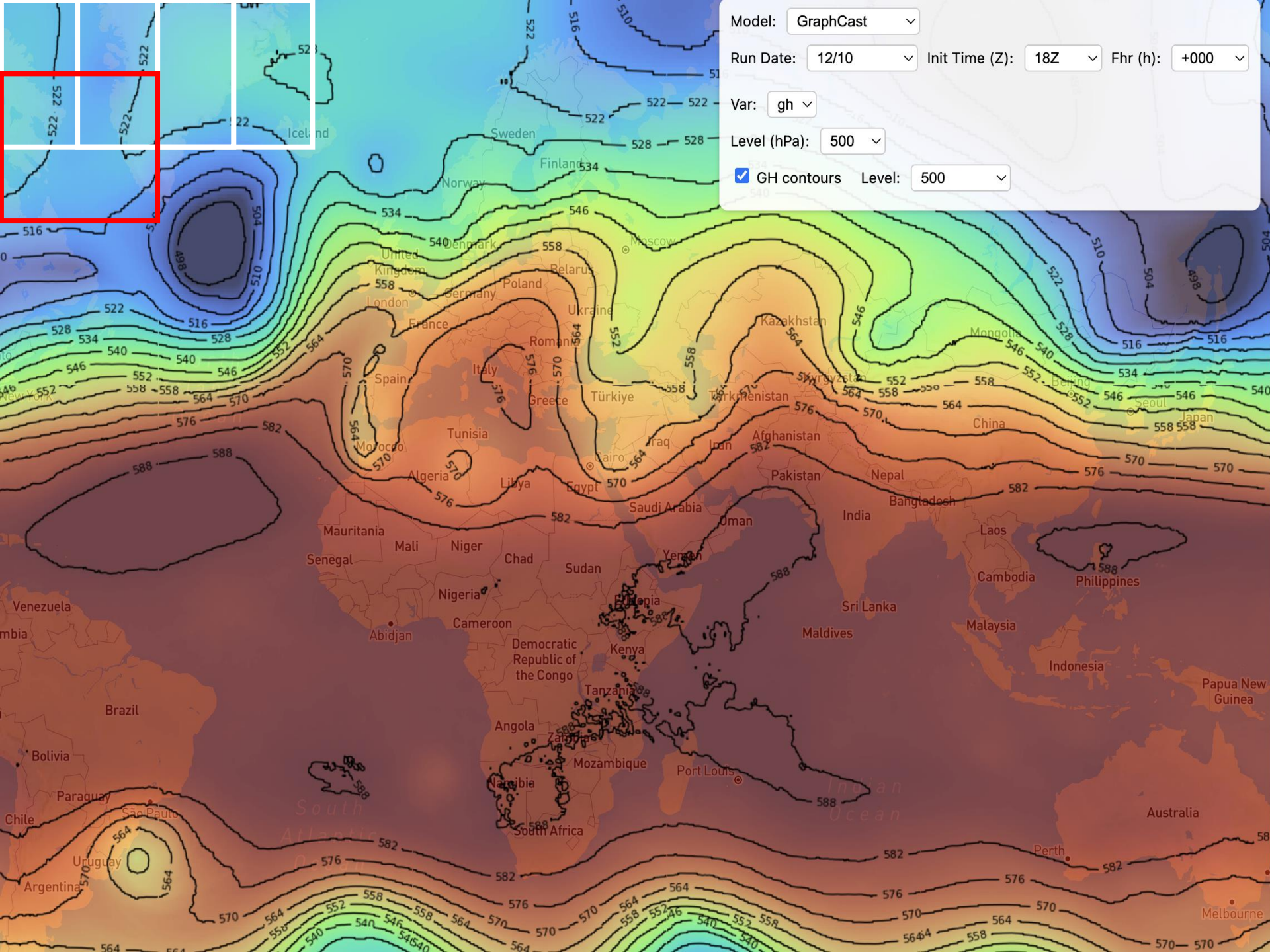
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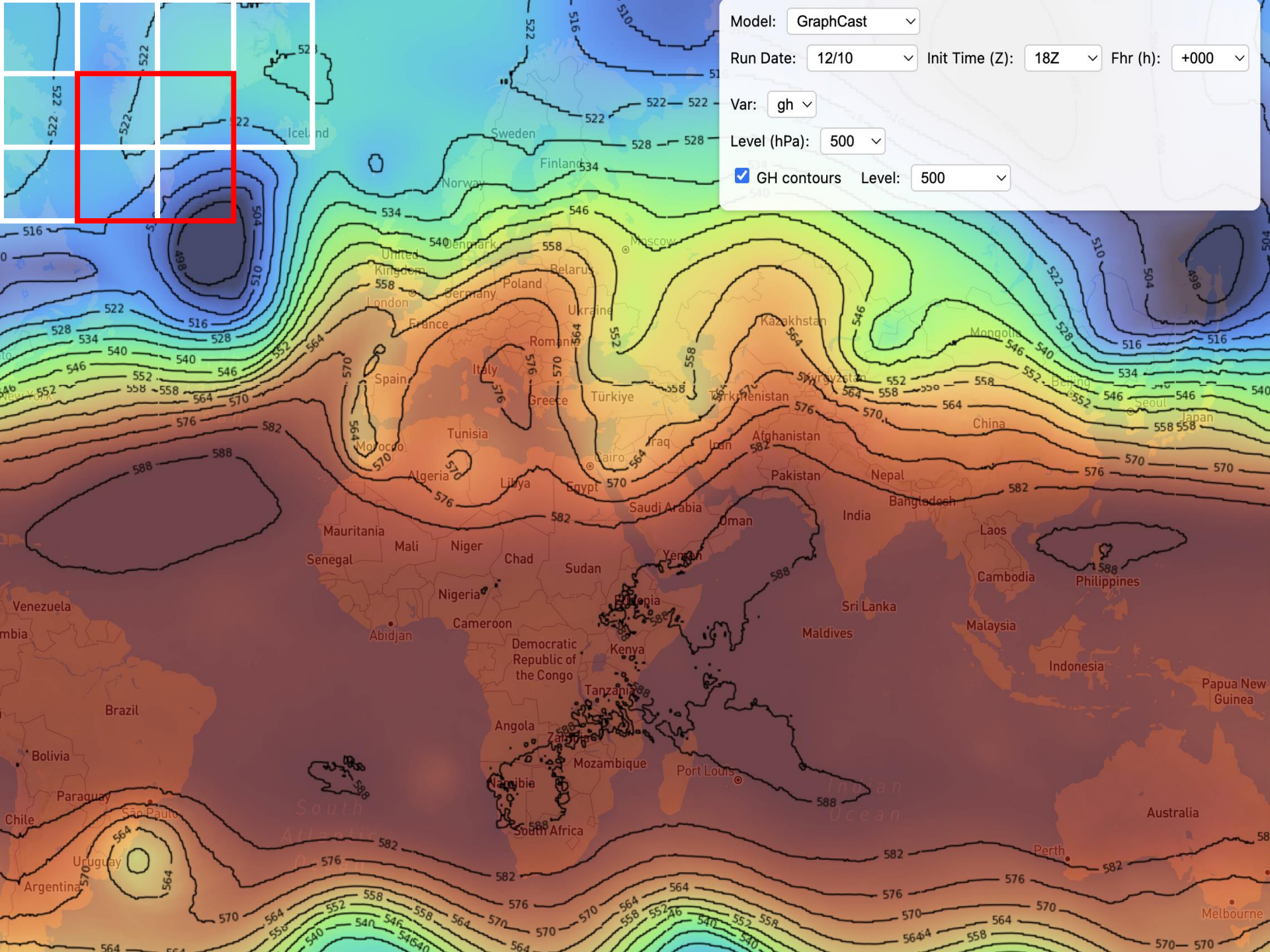
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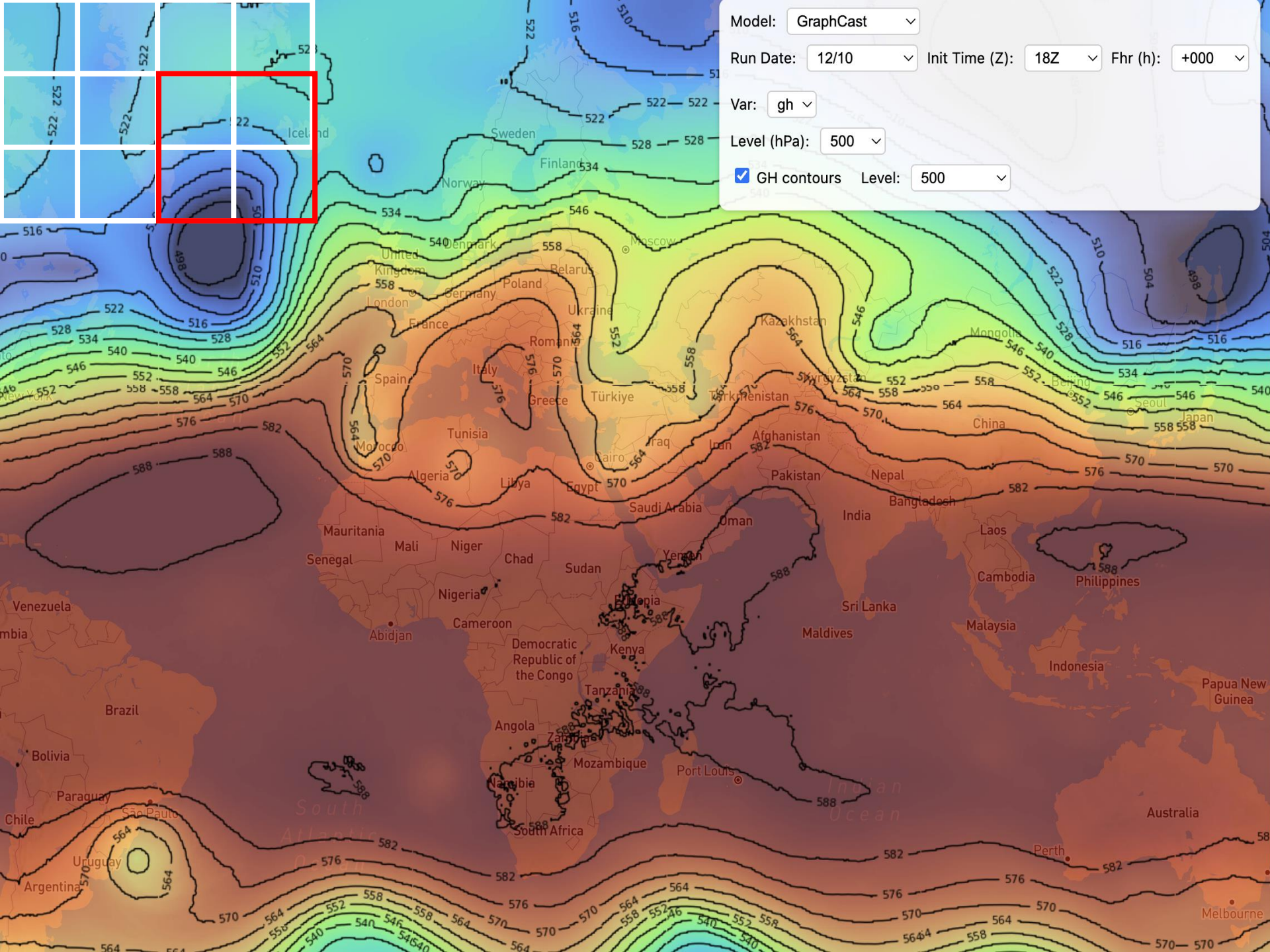
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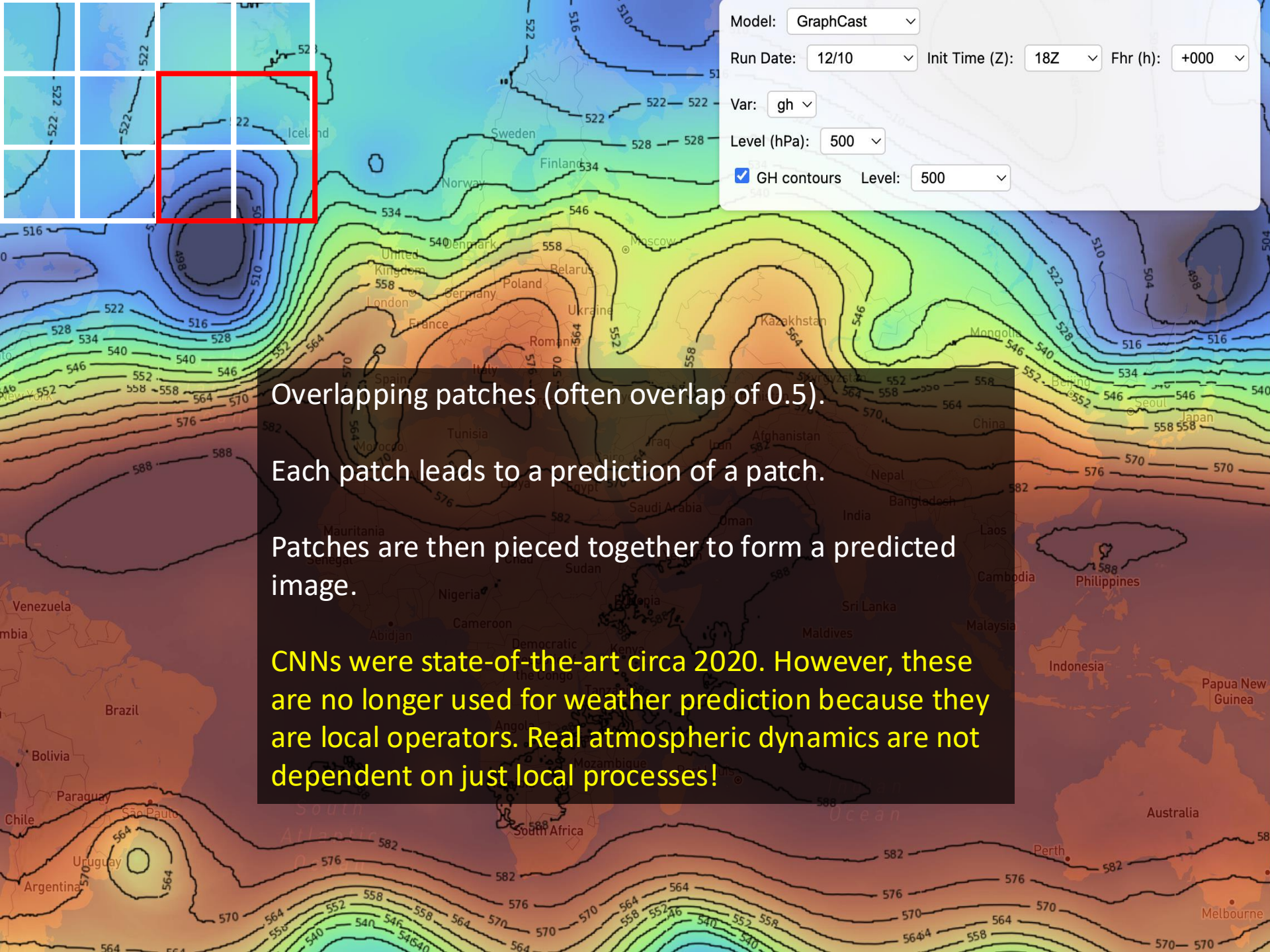


Overlapping patches (often overlap of 0.5).

Each patch leads to a prediction of a patch.

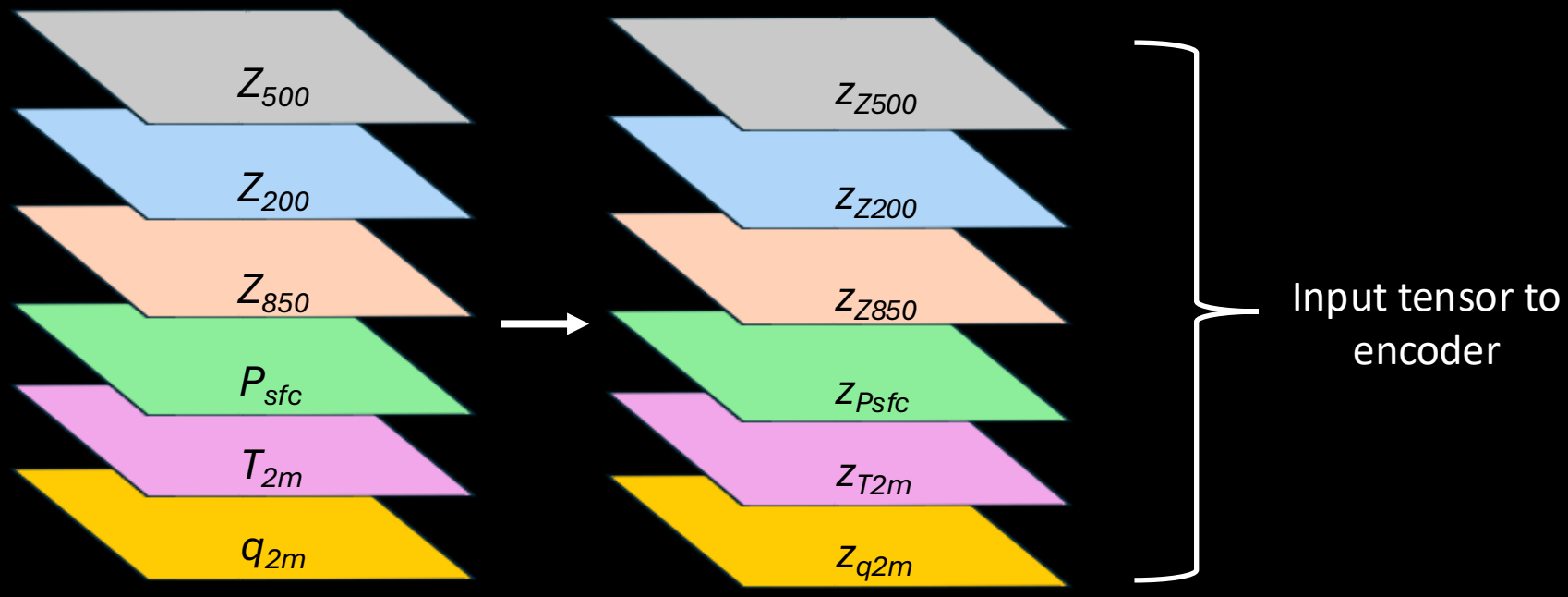
Patches are then pieced together to form a predicted image.

CNNs were state-of-the-art circa 2020. However, these are no longer used for weather prediction because they are local operators. Real atmospheric dynamics are not dependent on just local processes!

How do we use available data to train ML models of the atmosphere?

z-score normalization

$$z = (X - \mu)/\sigma$$



Raw data fields (These do not go into training!)

Normalized fields (mean = 0; stdev = 1)

Is my model still learning?

Do I need to adjust my learning rate?

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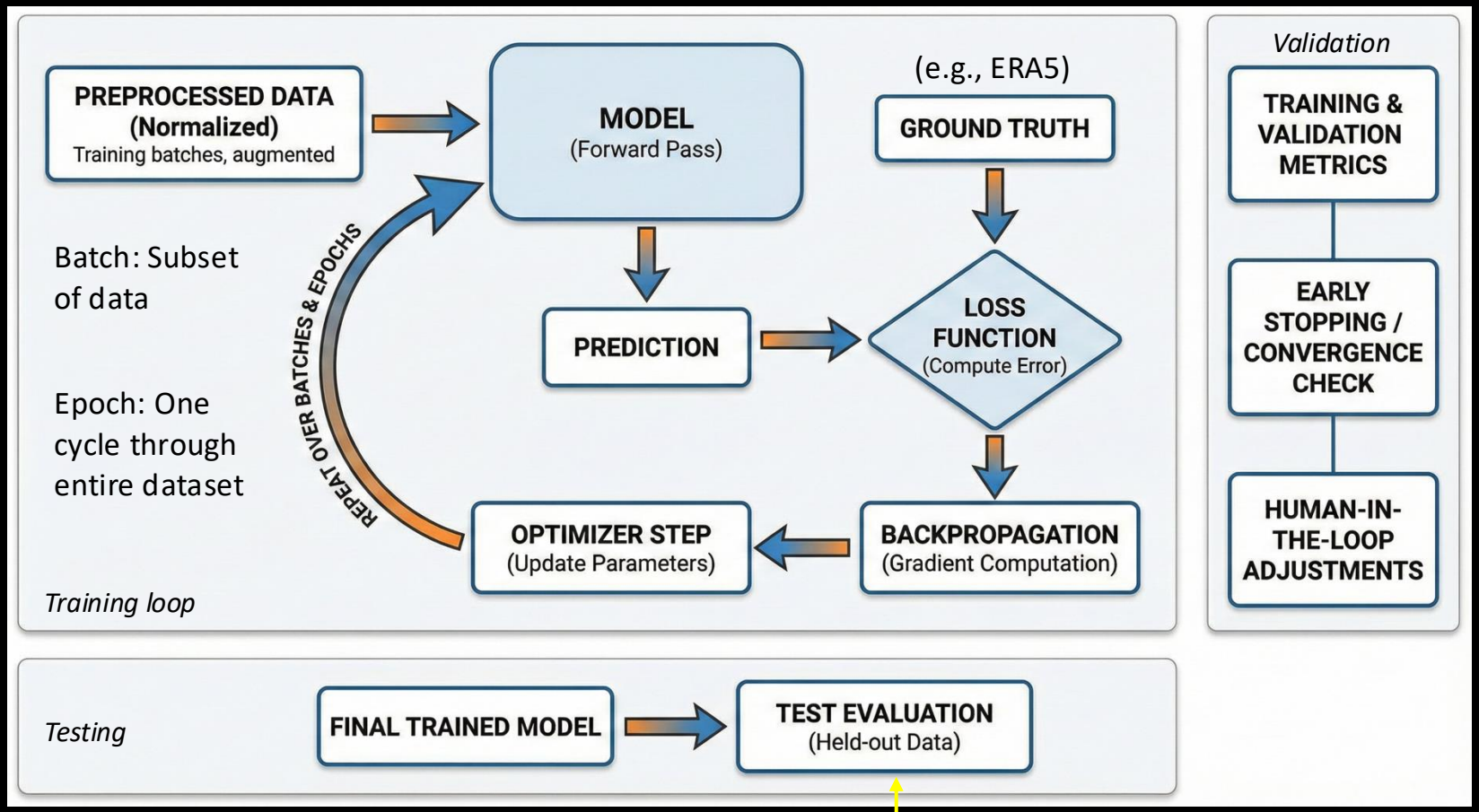
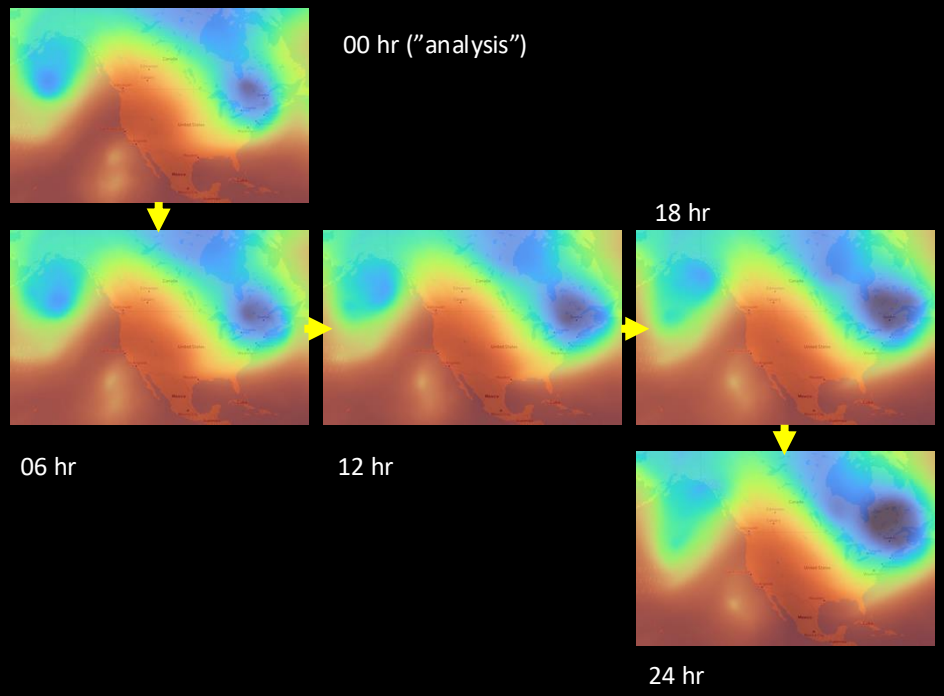


This data **MUST** be left out of training! This is required to show the model performs well on unseen data.

Error in ML Models

ML models don't integrate equations forward in time. But they still have similar limits of predictability. What produces the error?



Not propagating errors mathematically.

But there is still information content error in autoregressive models.

06 hr is IC for 12 hr.
12 hr is IC for 18 hr…and so on…

So any error in the 12 hr prediction is an error in the input tensor for the 18 hr prediction. Eventually, prediction does not match the truth.

Error in ML Models

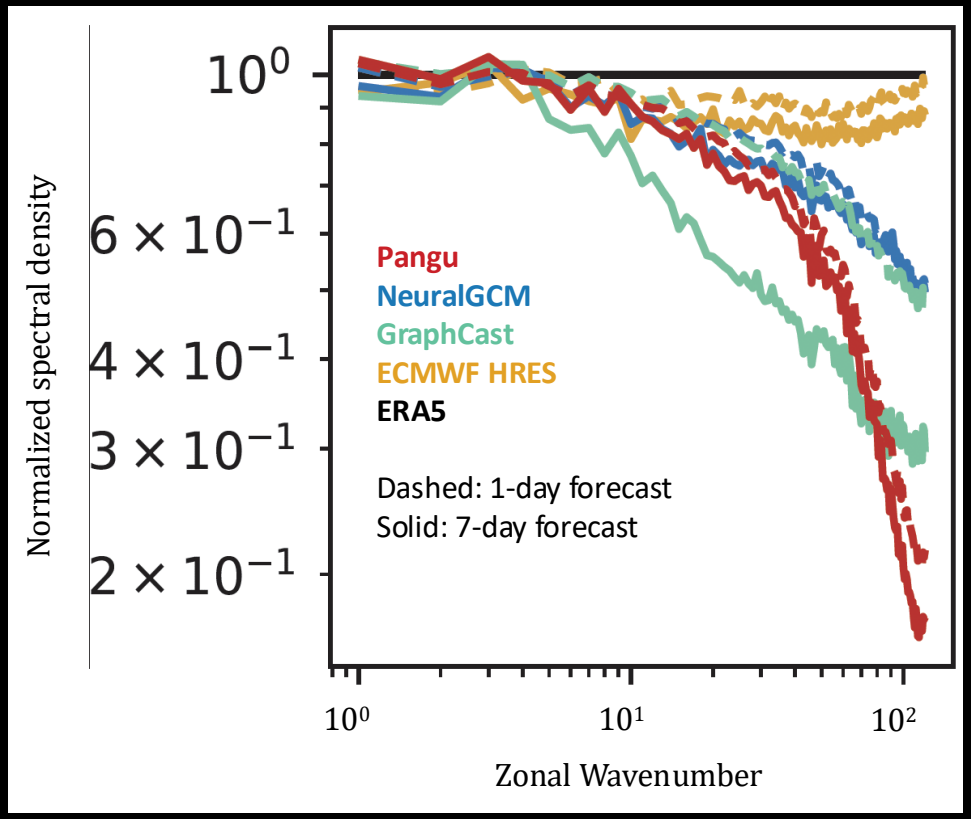
Smoothing

ML models universally underrepresent small scale weather.

Why? ML models aren't trained to obey physics! They are trained to minimize a loss function.

Huge impacts for real day-to-day weather prediction (temperature, rainfall, wind, etc.).

Normalized power spectra for wind speed in 15°S to 15°N



Error in ML Models

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Mean squared error (MSE) is often used as a loss function.

Truth

Prediction

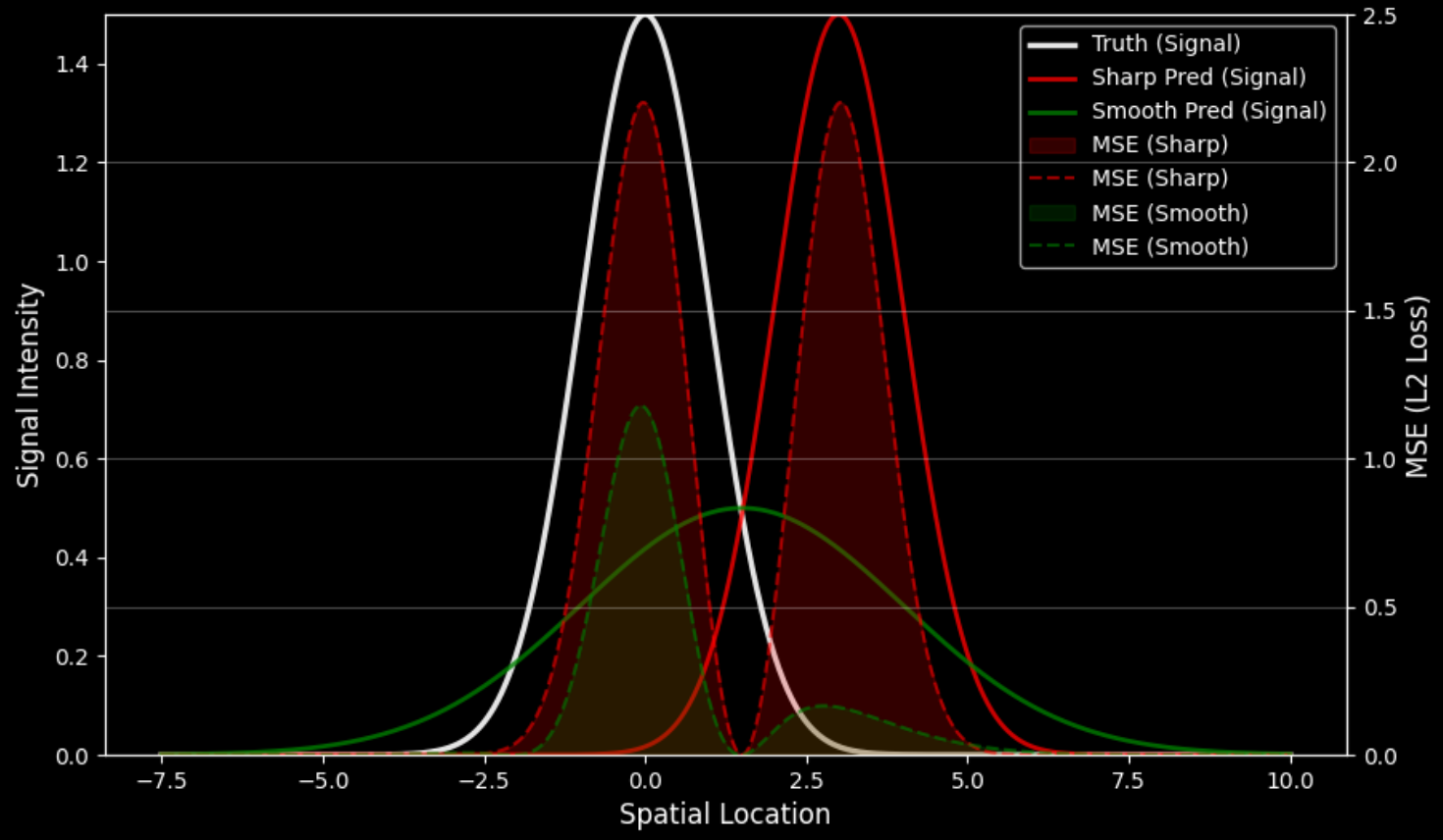
$$L = \frac{1}{N}\sum_{i=1}^{N}(x_i - \hat{x}_i)^2$$

Large errors are amplified!

Therefore, models learn to "play it safe" when trained. This causes them to lose vital information where a weather forecaster might care the most (where sharp gradients exist).

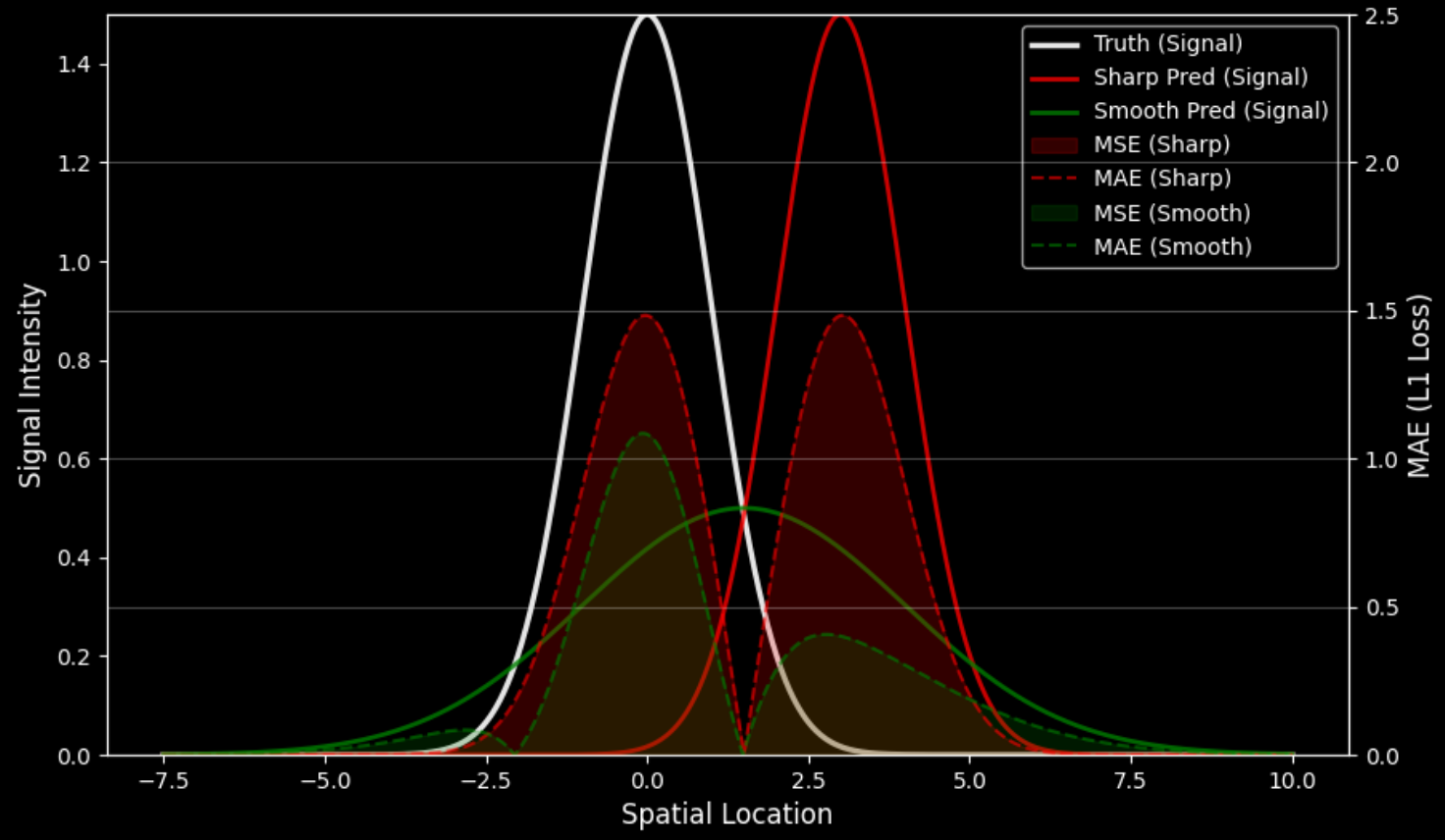
Example of mean squared error (MSE)

Mean L2 Loss (sharp): 0.41
Mean L2 loss (smooth): 0.12



Example of mean absolute error (MAE)

Mean L1 Loss (sharp): 0.37
Mean L1 loss (smooth): 0.21

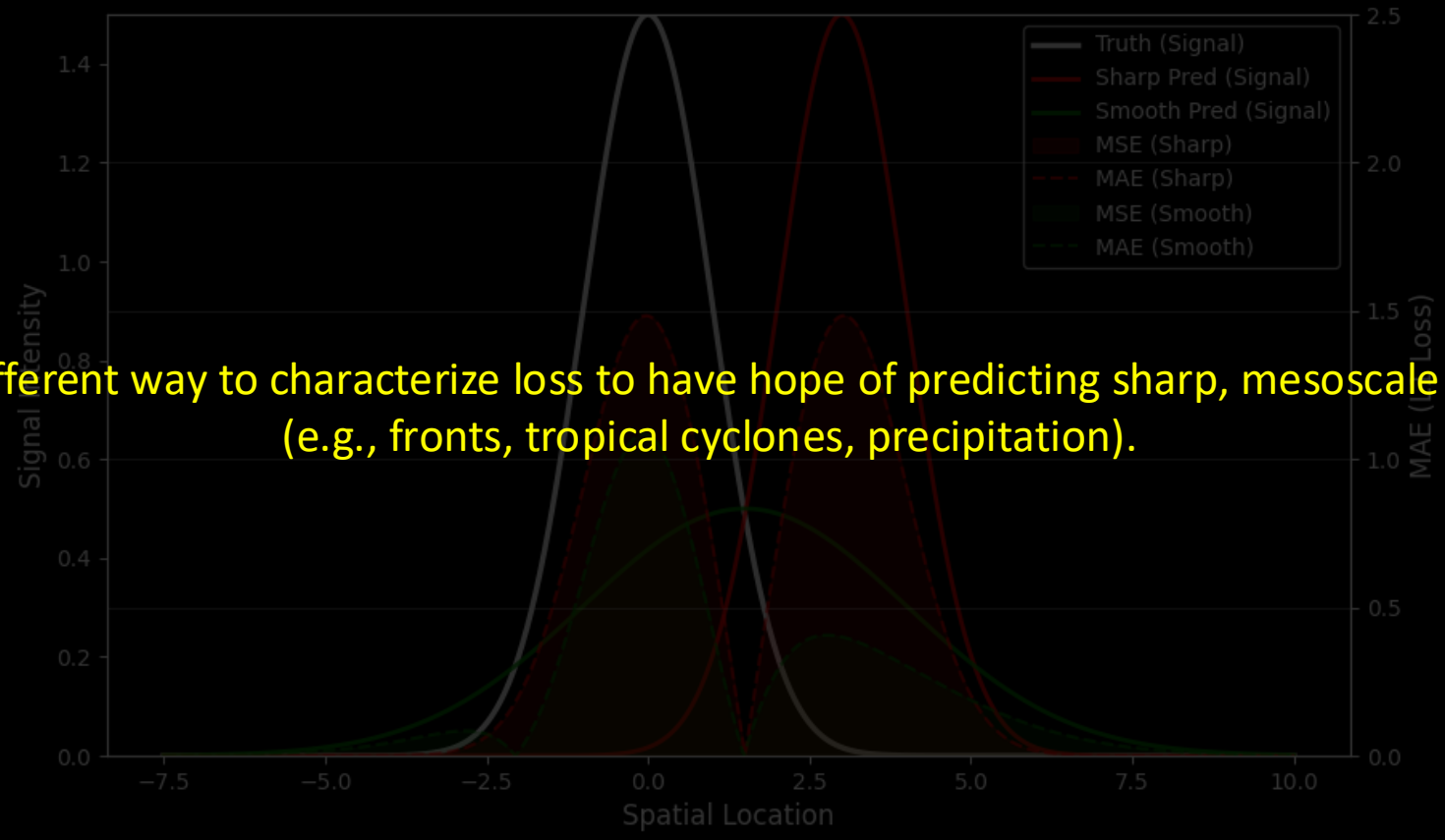


For high amplitude signals, L2 loss gets amplified much faster than L1 loss.

➡ *Spiky signals like heavy rain*

Example of mean absolute error (MAE)

Mean L1 Loss (sharp): 0.37
Mean L1 loss (smooth): 0.21



Need a different way to characterize loss to have hope of predicting sharp, mesoscale features (e.g., fronts, tropical cyclones, precipitation).

For high amplitude signals, L2 loss gets amplified much faster than L1 loss.

→ *Spiky signals like heavy rain*